PAPER_343 SGR J1745-2900: SC_m Mass-Modified Luminosity and Doubled f_react (June 2013)

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 96
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST UQFF treatment of magnetar SC_m mass modification with L_X = ?_vacf_resV
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Abstract

A novel UQFF form for the superconductive modifier SC_m of SGR J1745-2900 is derived as a mass-dependent suppression by the critical field ratio: SC_m = M(1 - B/B_crit). The X-ray luminosity is expressed as L_X = ?_vacf_resV, coupling vacuum energy density, resonance frequency, and magnetospheric volume. The activation event of June 2013 corresponds to a doubling of f_react, confirmed by the sudden spin-up and luminosity jump. T_surf = 1.16×107 K is derived from the Stefan-Boltzmann radiative balance.

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2. Core Physics

2.1 Mass-Modified Superconductive Modifier

$$\mathrm{SC}_m = M \cdot \left(1 - \frac{B}{B_{\mathrm{crit}}}\right)$$

where B_crit = 4.4×10 T (quantum critical field). For SGR J1745-2900: B = 2×10 T – B_crit, giving SC_m – M (nearly full superconductive coupling).

2.2 Vacuum-Energy X-ray Luminosity Form

$$L_X = \rho_{\mathrm{vac}} \cdot f_{\mathrm{res}} \cdot V_{\mathrm{mag}}$$

where ?_vac = ?_SCm - ?_UA is the net vacuum energy density and V_mag is the magnetospheric volume.

2.3 June 2013 Activation Event

$$f_{\mathrm{react}}^{\mathrm{post}} = 2 \cdot f_{\mathrm{react}}^{\mathrm{pre}}$$

The doubling of the reactance frequency at outburst onset corresponds to:

$$\Delta L_X = \rho_{\mathrm{vac}} \cdot \Delta f_{\mathrm{res}} \cdot V_{\mathrm{mag}} = \rho_{\mathrm{vac}} \cdot f_{\mathrm{react}}^{\mathrm{pre}} \cdot V_{\mathrm{mag}}$$

2.4 Surface Temperature

From Stefan-Boltzmann balance of magnetospheric luminosity:

$$T_{\mathrm{surf}} = \left(\frac{L_X}{4\pi R_{\mathrm{NS}}^2 \sigma_{\mathrm{SB}}}\right)^{1/4} = 1.16 \times 10^7 \ \mathrm{K}$$

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3. Key Values

Quantity	Formula	Value
SC_m	M(1-B/B_crit)	M_NS (B – B_crit)
L_X	?_vacf_resV	~105 erg/s
f_react (pre-2013)	canonical	f0
f_react (post June 2013)	2f0	2f0
T_surf	Stefan-Boltzmann	1.16×107 K
B_crit	Quantum critical	4.4×10 T

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4. Physical Significance

SGR J1745-2900 is the only magnetar within 0.3 pc of Sgr A*, making it the unique test-bed for UQFF near the Galactic Center supermassive black hole. The SC_m = M(1-B/B_crit) form establishes that even strongly magnetized neutron stars maintain near-unity superconductive coupling due to B – B_crit. The L_X = ?_vacf_resV form is a direct observable prediction: doubling f_res (as seen in June 2013) should produce a factor of 2 luminosity jump, consistent with the 2013 XMM-Newton observations.

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5. Deduplication Note

- **vs. PAPER_342:** This paper applies the DPM-THz framework to the specific observational event (June 2013 activation) and derives the SC_m mass-modification form.
- **vs. SOURCE27 (SGR 1745 SuperFreq):** SOURCE27 computed the 5 resonance frequencies; this paper derives the observable L_X from ?_vacf_resV.

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6. Classification

Physics Territory: FIRST UQFF SC_m mass-modified form for magnetar near Galactic Center
Scale: Stellar/Galactic Center (0.3 pc)
CP Implementation: `SGR17452900SCmLxFreqFormCalculator` (CondensedPhysics3.py, Session 96)

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]exp(-??t) = 1 - 5.7e-1 exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s.